

Project Executable Files

Date	29 June 2026
Team ID	XXXXXX
Project Name	OptiCrop Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Project Executable Files

This section documents all executable and deployable files for the OptiCrop Smart Agricultural Production Optimization Engine. The evaluator can run the application locally using the instructions below. No cloud deployment is required — the trained model artifacts are included in the repository so the application runs immediately after cloning.

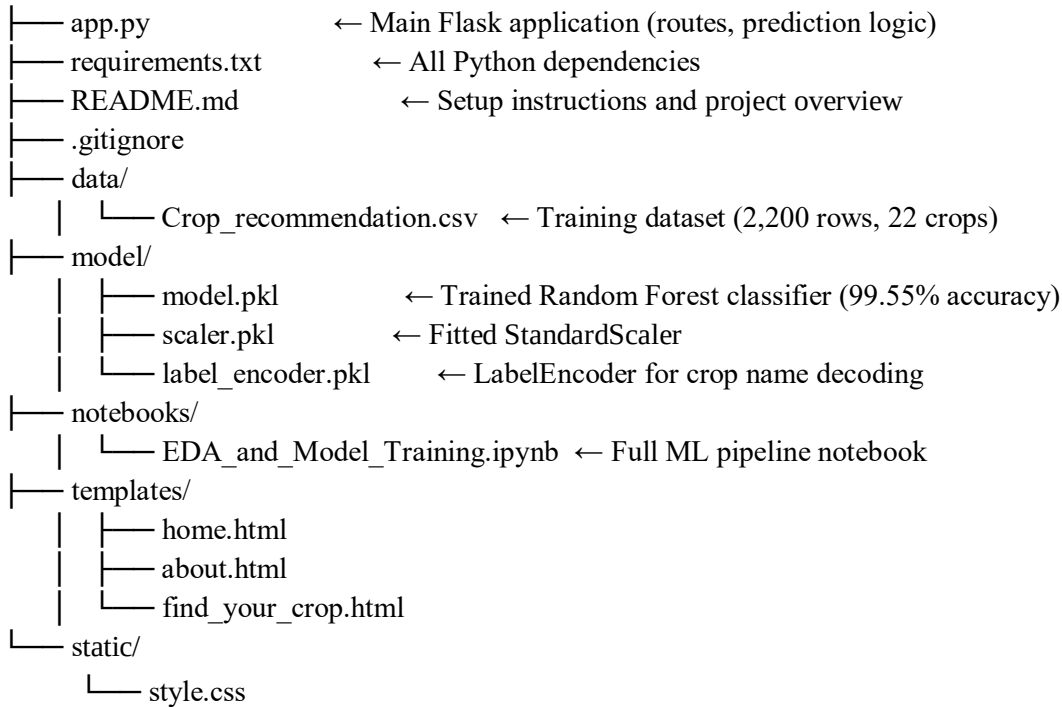
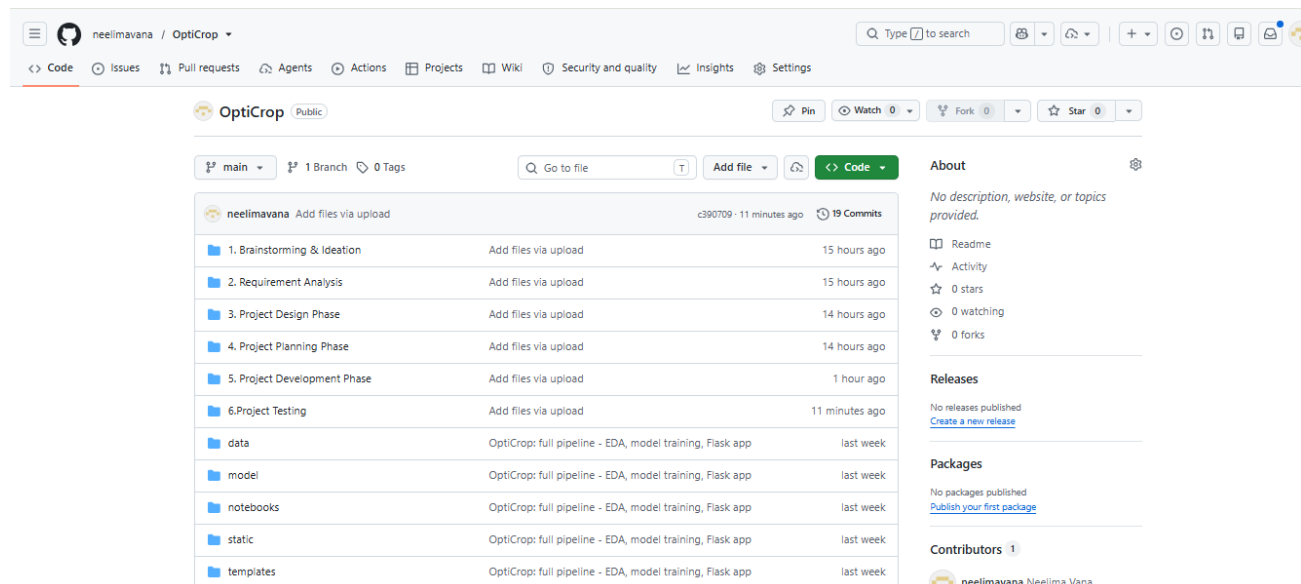
Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders) — app.py, templates/, static/, model/, notebooks/, data/	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt — lists all Python dependencies (numpy, pandas, scikit-learn, matplotlib, seaborn, scipy, flask, joblib)	Yes
4	Database schema / seed files (if applicable)	NA
5	Environment configuration file (.env.example or similar)	NA
6	Deployed application URL (if hosted) — application runs locally at http://127.0.0.1:5000	NA
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results — Performance Testing conducted using Apache JMeter 5.6.3	Yes
10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure

The OptiCrop repository is organized as follows:

OptiCrop/

The screenshot shows the GitHub repository page for 'neelimavana / OptiCrop'. The repository is public and has 19 commits. The file browser shows the following structure:

File/Folder	Description	Last Commit
1. Brainstorming & Ideation	Add files via upload	15 hours ago
2. Requirement Analysis	Add files via upload	15 hours ago
3. Project Design Phase	Add files via upload	14 hours ago
4. Project Planning Phase	Add files via upload	14 hours ago
5. Project Development Phase	Add files via upload	1 hour ago
6. Project Testing	Add files via upload	11 minutes ago
data	OptiCrop: full pipeline - EDA, model training, Flask app	last week
model	OptiCrop: full pipeline - EDA, model training, Flask app	last week
notebooks	OptiCrop: full pipeline - EDA, model training, Flask app	last week
static	OptiCrop: full pipeline - EDA, model training, Flask app	last week
templates	OptiCrop: full pipeline - EDA, model training, Flask app	last week

The right sidebar shows the repository's metadata:

- About:** No description, website, or topics provided.
- Readme:** No description provided.
- Activity:** 0 stars, 0 watching, 0 forks.
- Releases:** No releases published. [Create a new release](#)
- Packages:** No packages published. [Publish your first package](#)
- Contributors:** 1 contributor: neelimavana (Neelima Vana)

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	Not hosted — runs locally at http://127.0.0.1:5000
Login Credentials (Demo)	No login required — application is open access. Navigate directly to http://127.0.0.1:5000 to access the Home page.
Platform / Hosting Provider	Local — Flask development server on Windows 11
Repository Link	https://github.com/neelimavana/OptiCrop
Demo Video Link	[Insert YouTube/Google Drive demo video link here]

Step 4: Run Instructions

Follow these steps to run the OptiCrop application locally:

Step 1. Clone the repository:

```
git clone https://github.com/neelimavana/OptiCrop.git
```

Step 2. Navigate into the project folder:

```
cd OptiCrop
```

Step 3. Create a virtual environment:

```
python -m venv venv
```

Step 4. Activate the virtual environment (Windows):

```
venv\Scripts\activate
```

Step 5. Install all dependencies:

```
pip install -r requirements.txt
```

Step 6. Run the Flask application:

```
python app.py
```

Step 7. Open your browser and go to:

```
http://127.0.0.1:5000
```

Step 8. Navigate to 'Find Your Crop' → enter soil and climate values → click Predict.

Note: The trained model files (model.pkl, scaler.pkl, label_encoder.pkl) are already included in the model/ folder. No retraining is needed — the application runs immediately after Step 6.

```
warnings.warn(
127.0.0.1 - - [05/Jul/2026 13:42:02] "POST /find-your-crop HTTP/1.1" 200 -
(.venv) PS C:\Users\ADMIN\OptiCrop> python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
```

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	No user authentication — any user can access the prediction form without logging in.	Planned for future scope — user login/registration with session management.
2	Application runs only on local server — no cloud deployment currently available.	Can be deployed to Render or Railway with minimal configuration changes. Noted as future enhancement.
3	Model trained on a fixed dataset of 22 crops — may not cover all regional crop varieties in India.	Future scope includes training on a larger, regionally-specific dataset covering more crop types.